

Multi-Center A_1 Mixtures at the Rubio de Francia Endpoint

Run fa_banach_001, agent_lane_16

2026-08-11

Source question and status

For a pairwise disjoint family of intervals Ω in $\mathbb{R}$, let

$$T^\Omega f(x) = \left(\sum_{\omega \in \Omega} |H_\omega f(x)|^2 \right)^{1/2}, \quad \widehat{H_\omega f} = \mathbf{1}_\omega \widehat{f}.$$

After Theorem 5.2 of Amenta–Lorist–Veraar, arXiv:1703.06044, the strong estimate at the scalar endpoint $p = q'$ is recorded as open. Its central $q = 2$ case asks whether

$$\|T^\Omega f\|_{L^2(w)} \lesssim_w \|f\|_{L^2(w)} \quad \text{for every } w \in A_1, \quad (1)$$

uniformly in Ω . Di Plinio–Flórez–Amatriain–Parissis–Roncal, arXiv:2308.01442, still call (1) the outstanding trigonometric endpoint conjecture. They prove it when w is even and radially decreasing, as well as for the Walsh analogue.

The result below closes their radial class under arbitrary positive mixtures. It produces multi-center and multiscale weights which need not be radially decreasing about any point.

Proof intuition

The key observation is elementary but powerful: after squaring, an $L^2(w)$ estimate is linear in w . If the endpoint estimate holds with one common constant for each member of a family (v_s) , Tonelli’s theorem lets us integrate the estimates and proves the same bound for every positive mixture $w = \int v_s$. Uniform A_1 control is also preserved because the maximal operator is sublinear. Combining this cone principle with the 2023 radial theorem gives the claimed class.

Lemma 1 (positive-mixture principle). *Let (S, ν) be a measure space and let v_s be nonnegative locally integrable weights, measurable in (s, x) . Suppose that, with a common constant B ,*

$$\|T^\Omega f\|_{L^2(v_s)} \leq B \|f\|_{L^2(v_s)}$$

for ν -almost every s , every finite pairwise disjoint Ω , and every Schwartz function f . If $w(x) = \int_S v_s(x) d\nu(s)$ is finite almost everywhere and locally integrable, then

$$\|T^\Omega f\|_{L^2(w)} \leq B \|f\|_{L^2(w)}.$$

If in addition $[v_s]_{A_1} \leq K$ almost everywhere, then $[w]_{A_1} \leq K$.

Proof. Tonelli's theorem gives

$$\int_{\mathbb{R}} |T^{\Omega} f|^2 w = \int_S \int_{\mathbb{R}} |T^{\Omega} f|^2 v_s dx d\nu(s) \leq B^2 \int_S \int_{\mathbb{R}} |f|^2 v_s dx d\nu(s) = B^2 \int_{\mathbb{R}} |f|^2 w.$$

For the last assertion, positivity and sublinearity give, almost everywhere,

$$\mathcal{M}w \leq \int_S \mathcal{M}v_s d\nu(s) \leq K \int_S v_s d\nu(s) = Kw.$$

□

Theorem 1 (multi-center radial mixtures). *Fix $K < \infty$. For $s \in S$, let*

$$v_s(x) = \phi_s\left(\frac{|x - a_s|}{r_s}\right), \quad a_s \in \mathbb{R}, \quad r_s > 0,$$

where $\phi_s : [0, \infty) \rightarrow [0, \infty)$ is nonincreasing and $[v_s]_{A_1} \leq K$. Let $w = \int_S v_s d\nu(s)$ be finite almost everywhere and locally integrable. There is $C(K) < \infty$, independent of the mixture and of the disjoint family Ω , such that

$$\|T^{\Omega} f\|_{L^2(w)} \leq C(K) \|f\|_{L^2(w)}.$$

Moreover $w \in A_1$ and $[w]_{A_1} \leq K$.

Proof. Translation and dilation preserve both the A_1 characteristic and the form of the Fourier-projection inequality. Thus the radial endpoint theorem of arXiv:2308.01442 applies to every v_s , with a bound depending only on K (its stated bound involves $[v_s]_{A_1}$ and $[v_s]_{A_{\infty}}$, and the latter is controlled by the former). Lemma 1 finishes the proof. □

Here is a concrete, flexible subclass.

Proposition 1 (polynomial-tail mixtures). *Fix $0 < \alpha < 1$. Let $c_j \geq 0$, $a_j \in \mathbb{R}$, and $r_j > 0$, and assume that*

$$w(x) = \sum_j c_j \left(1 + \frac{|x - a_j|}{r_j}\right)^{-\alpha}$$

is finite almost everywhere and locally integrable. Then $w \in A_1$, with $[w]_{A_1} \leq K_{\alpha}$, and

$$\|T^{\Omega} f\|_{L^2(w)} \leq C_{\alpha} \|f\|_{L^2(w)}$$

uniformly over all pairwise disjoint Ω . The same conclusion holds for positive integral mixtures of these kernels.

Proof. It suffices to check that $k(x) = (1 + |x|)^{-\alpha}$ has an A_1 constant depending only on α . Fix an interval $I \ni x$, put $R = 1 + |x|$, and let $L = |I|$. If $L \leq R/2$, then $1 + |y| \geq R/2$ for $y \in I$, so the average of k on I is at most $2^{\alpha} k(x)$. If $L > R/2$, then

$$\frac{1}{L} \int_I (1 + |y|)^{-\alpha} dy \leq C_{\alpha} L^{-\alpha} \leq C'_{\alpha} R^{-\alpha} = C'_{\alpha} k(x).$$

Translation and dilation do not change this bound. The result follows from Theorem 1; integral mixtures follow from Tonelli. □

For example, two well-separated kernels in Proposition 1 give two strict peaks, so the resulting weight is not radially decreasing about any center. Arbitrarily many centers and scales are allowed, and the constant does not grow with their number.

A reduction toward the full conjecture

The cone principle also narrows the remaining problem. Write $\mathcal{M}$ for the uncentered Hardy–Littlewood maximal operator.

Proposition 2 (Coifman–Rochberg reduction). *Fix K . There exist $r > 1$ and $\delta \in (1/r, 1)$, depending only on K , with the following property. Every $w \in A_1$ with $[w]_{A_1} \leq K$ is pointwise comparable, with constants depending only on K , to a positive mixture*

$$v = \sum_{k \in \mathbb{Z}} 2^k (\mathcal{M} \mathbf{1}_{E_k})^\delta, \quad E_k = \{x : w(x) > 2^k\}.$$

Each $(\mathcal{M} \mathbf{1}_{E_k})^\delta$ has a common A_1 bound. Consequently, to prove (1) for all weights of A_1 characteristic at most K , it is enough to prove it uniformly for the special weights $(\mathcal{M} \mathbf{1}_E)^\delta$, for arbitrary measurable $E \subset \mathbb{R}$.

Proof. The reverse-Hölder theorem for A_1 gives $r > 1$ and $A = A(K)$ such that

$$\frac{1}{|I|} \int_I w^r \leq A \left(\frac{1}{|I|} \int_I w \right)^r.$$

At almost every x , for every interval $I \ni x$, Chebyshev and the A_1 inequality imply

$$\frac{|E_k \cap I|}{|I|} \leq 2^{-kr} \frac{1}{|I|} \int_I w^r \leq AK^r \left(\frac{w(x)}{2^k} \right)^r.$$

Taking the supremum over I yields

$$\mathcal{M} \mathbf{1}_{E_k}(x) \leq \min \left\{ 1, AK^r \left(\frac{w(x)}{2^k} \right)^r \right\}. \quad (2)$$

Choose $\delta \in (1/r, 1)$. At Lebesgue points in E_k , $\mathcal{M} \mathbf{1}_{E_k} = 1$. Hence the terms with $2^k < w(x)$ sum to a quantity comparable from above and below to $w(x)$. For the remaining terms, (2) and $r\delta > 1$ give a convergent geometric series:

$$\sum_{2^k \geq w(x)} 2^k (\mathcal{M} \mathbf{1}_{E_k}(x))^\delta \leq C(K) w(x)^{r\delta} \sum_{2^k \geq w(x)} 2^{-k(r\delta-1)} \leq C'(K) w(x).$$

Thus $w \lesssim v \lesssim_K w$. The standard Coifman–Rochberg lemma states that $(\mathcal{M}g)^\delta \in A_1$, with constant depending only on δ , for every nonnegative g . If the desired endpoint estimate holds uniformly for these special weights, Lemma 1 proves it for v ; pointwise comparability transfers it from v to w . $\square$

Remark 1 (scope). *Theorem 1 is an unconditional partial result. Proposition 2 is a reduction, not a proof of the full conjecture: a general weight $(\mathcal{M} \mathbf{1}_E)^\delta$ need not be a positive mixture of single-center radial weights by any argument presently supplied here. The case of all A_1 weights, and the source’s other endpoint cases $p = q' < 2$, remain open.*

Novelty check

Exact-id, title, endpoint-phrase, $p = q'$, A_1 , radial-weight, Walsh, and positive-mixture searches were run in the local FA/Banach registry and in primary arXiv sources through 2026-08-11. The decisive later source was arXiv:2308.01442. No full endpoint resolution and no explicit statement of the multi-center positive-mixture consequence above was found. Since the main proof is a short cone argument from the 2023 theorem, novelty confidence is moderate while validity confidence is high.

References

- A. Amenta, E. Lorist, and M. Veraar, *Rescaled extrapolation for vector-valued functions*, arXiv:1703.06044.
- F. Di Plinio, M. Flórez-Amatriain, I. Parissis, and L. Roncal, *Pointwise localization and sharp weighted bounds for Rubio de Francia square functions*, arXiv:2308.01442.
- R. R. Coifman and R. Rochberg, *Another characterization of BMO*, Proc. Amer. Math. Soc. 79 (1980), 249–254 (maximal-function powers in A_1).